



Building AI Agents

What This Session Is NOT

Not ML training

Not prompt engineering poetry

Not AGI

Not "ChatGPT as an intern"

Not production-ready magic

What an LLM Actually Is

A frozen statistical model

Input tokens → probability distribution via LARGE matrix multiplications → output tokens

No memory

No learning during usage

No intent or goals

Myths

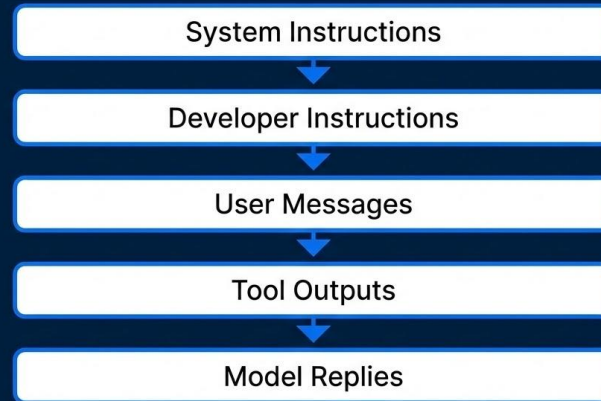
- The Model Learns From My Prompt
 - Weights are frozen
 - Each request starts from the same base model
- Local LLMs Call OpenAI
 - Models run fully locally
- LLMs Think in English
 - The model operates on tokens i.e. floats
- JSON Output Is Guaranteed
 - Instructions are soft constraints
- Guardrails Prevent Prompt Injection
 - Guardrails are advisory
 - Untrusted text in context = risk
- LLM Output Is Deterministic
 - Sampling introduces randomness
 - Different models $\neq$ same answers
- The LLM Will Figure It Out
 - Ambiguity produces confident nonsense
 - LLMs optimize for plausibility, not truth
 - Clarity is the developer's job

What Is Context?

Context is everything the model sees every time

- Context contains the conversation between user and Llm as a list
- Context window is finite, old tokens are silently dropped
- The model does not know what it forgot

Generative AI Model Inputs and Outputs



Tokens

- Tokens $\neq$ words
- Tokens are numerical representations of words
- Code and JSON are token-heavy
- Logs are context poison
- English: 100 words $\approx$ 130 tokens or 0.75 words per token
- Hindi: 100 words $\approx$ 170–200 tokens or 1.7–2 tokens per word

Chatbot vs Agent

Chatbot

- Input → output
- Stateless
- No tools

Agent

- Goal-driven loop
- State
- Tools
- Decision making

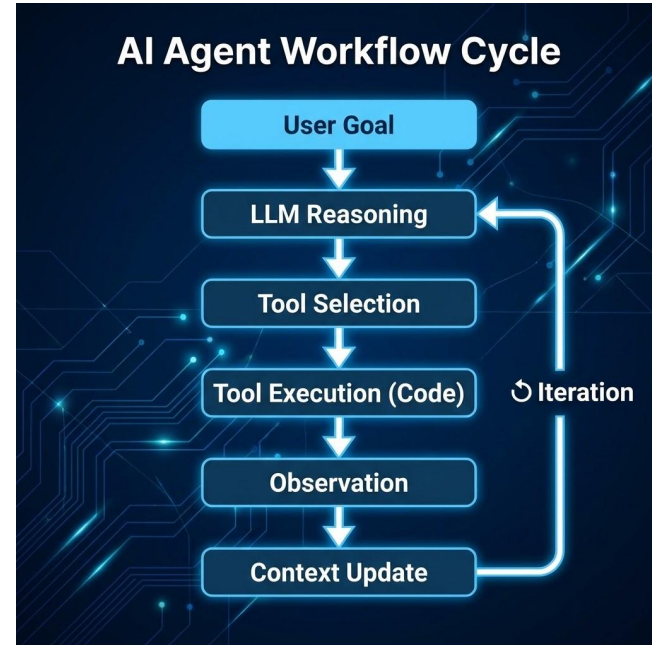
An agent is a loop around an LLM

Core Components of an AI Agent

- LLM (reasoning engine)
- Context (working memory)
- Tools (capabilities)
- State (what happened)
- Policy (when to stop)

No component is optional

- LLM proposes a structured tool call
- Your code executes it
- Output is injected back into context



Demo